

A short introduction to AUTOSTRUCTURE

AUTOSTRUCTURE is a suite of atomic-structure codes that is particularly geared towards the calculation of recombination cross sections and rate coefficients. It has been coded in the programming language FORTRAN by Nigel Badnell [1] from the University of Strathclyde (Glasgow, UK) and is available from a server there (<http://www.apap-network.org/autos/>). The code consists of the atomic-structure code `asdeck.f` and various post-processors for different type of data. For DR and RR calculations there are the post-processors `adasdr.f` and `adasrr.f`. Documentation is available as <http://www.apap-network.org/autos/ver/WRITEUP>.

1 Installation

A very basic installation procedure is described in the document <http://www.apap-network.org/autos/ver/INSTALL>. In order to be able to more conveniently handle the various input and output files that are read and generated by the various codes, a makefile and several shell scripts have been developed. Just type "make" to see all installation options. The essential individual installation steps are:

- compilation of `asdeck.f`: "make AUTOS",
- compilation of `adasdr.f`: "make AUTOSDR",
- to create soft-links in the `~/bin` directory to the shell scripts: "make install",
- to make all shell scripts executable: "chmod 755 *.sh",
- to extract documentation from the source files: "make doc",
- to include the `~/bin` directory in your standard path: add the line
"export PATH=\$PATH:\$HOME/bin" to the file `.bashrc` in your home-directory.

2 DR calculations with AUTOSTRUCTURE

AUTOSTRUCTURE has to calculate all atomic level energies and transition rates which are required for the computation of DR cross sections. This is done partly by `asdeck.f` and partly (for high- n levels) by `adasdr.f`. The input files have to be named `<name>.in` and `<name>.drin`, respectively, where `<name>` can be chosen freely. All files pertaining to a given calculation are

either stored in an automatically created subdirectory <name> or are named <name>.<ext> with various extensions <ext> as detailed below. In the following, DR of Be-like Ar¹⁴⁺ is treated as an example.

2.1 Level energies

Level energies of the Ar¹⁴⁺ $1s^2 2s^2$, $1s^2 2s 2p$, and $1s^2 2p^2$ configurations.

Input file Ar14.in:

```
A.S. Ar14+ energy levels
&SALGEB CUP='IC' MXCONF=3 MXVORB=2 TITLE='Ar14'
      KCOR1=1 KCOR2=1 &END
2 0    2 1
2 0
1 1
0 2
&SMINIM NZION=-18 PRINT='FORM' &END
```

Comments:

- FORTRAN input can be made via so-called namelists which group variables together. In the above example there are the two namelists &SALGEB and &SMINIM. The namelist input is terminated with &END.
- The ion charge is given as NZION. The negative sign refers to the computational method for the evaluation of radial wavefunctions (see WRITEUP for details).
- The atomic configurations are specified by the range of fixed orbitals. Presently, only the 1s orbital is fixed (KCOR1=1, KCOR2=1). In addition there are MXVORB=2 orbitals, i.e., the 2s and 2p orbitals, which are specified as "2 0 2 1" on a separate line immediately below the namelist &SALGEB input.
- The subsequent MXCONF=3 lines specify the configurations "2 0" [(1s²) 2s² 2p⁰], "1 1" [(1s²) 2s¹ 2p¹], and "0 2" [(1s²) 2s⁰ 2p²].
- "CUP='IC'" selects the angular momentum coupling which is intermediate coupling in this case.
- "PRINT='FORM'" requests formatted human-readable output.

Executing the calculation: Type "autos Ar14" on the command line.

Output file Ar14/LEVELS:

2J	P	S	L	CF	NI	ENERGY (RYD)
0	0	1	0	1	1	0.00000000
0	1	-3	1	2	4	2.11717592
2	1	-3	1	2	2	2.18220913
4	1	-3	1	2	1	2.32660879
2	1	-1	1	2	3	4.22709811
0	0	3	1	3	4	5.56803877
2	0	3	1	3	3	5.65434218
4	0	3	1	3	2	5.77786583
4	0	1	2	3	1	6.39944195
0	0	1	0	3	5	7.78202330
0	0	0	0	0	0	-758.69761466

The level energies are in units of Rydbergs (1 Ryd = 13.606 eV).

2.2 First step of DR calculation

DR and TR of $\text{Ar}^{14+}(1s^2 2s^2)$ via $2s \rightarrow 2p$ and $2s^2 \rightarrow 2p^2$ excitations involving $1s^2 2s 2p n\ell$ and $1s^2 2p^2 n\ell$ intermediate configurations.

Input file Ar14DR.in:

```

A.S. DR + TR of Ar14+
&SALGEB RUN='DR' RAD='YES' CUP='IC' MXCONF=3 MXCCF=3 MXVORB=2
      TITLE='2-2,n' KCOR1=1 KCOR2=1 LCON=5 &END
2 0 2 1
2 0
1 1
0 2

2 1
1 2
0 3

```

```

&DRR NMIN=3 NMAX=30 LMIN=0 LMAX=18 &END
&SMINIM NZION=-18 PRINT='UNFORM' &END
&SRADCON MENG=-9 &END
0.00 8.00

```

Comments:

- Additional items are specified in the namelist &SALGEB. There are MXCFF=3 explicitly specified configurations of the recombined ion, i.e., "2 1" $[(1s^2) 2s^2 2p^1]$, "1 2" $[(1s^2) 2s^1 2p^2]$, and "0 3" $[(1s^2) 2s^0 2p^3]$. These are the final $(1s^2) 2s^2 2p^1$ configuration and the intermediate $1s^2 2p^2 n\ell$ configurations with $n = 2$.
- The other $1s^2 2p^2 n\ell$ intermediate configurations with $NMIN \leq n \leq NMAX$ and $LMIN \leq \ell \leq LMAX$ are specified in the namelist &DRR. Internally AUTOSTRUCTURE extends the calculations to higher n and ℓ values up to $n = 999$ using hydrogenic rates and an interpolation scheme, where actually only a limited set of n -values is explicitly calculated. If you really want to stop the calculation at $n = NMAX$, you need to add NMESH = 0 to the namelist &DRR.
- The continuum energies of the initially free electron are specified in the namelist &SRADCON and the line below. There are |MENG|=9 energies in the energy range 0–8 Ryd. The negative sign of MENG is essential as explained in the WRITEUP. The highest energy should be larger than the largest excitation energy in the file Ar14/LEVELS calculated in section 2.1.

Executing the calculation: Type "autos Ar14DR" on the command line.

Output files: The file Ar14DR/olu contains intermediate output which is to be processed in step 2 below. If something goes wrong, the file Ar14DR/olg may contain error messages.

2.3 Second step of DR calculation

Input file Ar14DR.drin:

```

/IC/
&ONE COREX='2-2' NTAR1=2 &END
&TWO NR1=3 NBIN=4001 EMIN=0.0 EMAX=8.0
EWIDTH=0.0 TPER=1.47e-3 TPAR=1.47e-5
NECOR=10 &END
0 0 1 0 1 1 0.00000000

```

```

0 1 -3 1 2 4 2.11717592
2 1 -3 1 2 2 2.18220913
4 1 -3 1 2 1 2.32660879
2 1 -1 1 2 3 4.22709811
0 0 3 1 3 4 5.56803877
2 0 3 1 3 3 5.65434218
4 0 3 1 3 2 5.77786583
4 0 1 2 3 1 6.39944195
0 0 1 0 3 5 7.78202330
0 0 0 0 0 0 -758.69761466
0.0 2.0839 2.14931612 2.30258551 4.1206 5.51241 5.60557
5.72556 6.39944195 7.6603

```

Comments:

- Cross sections will be calculated for "NBIN=4001" electron energies ranging from $E_{\text{MIN}}=0.0$ to $E_{\text{MAX}}=8.0$ Ryd.
- $E_{\text{WIDTH}}=0.0$ flags the calculation of merged-beams rate coefficients (cf. Eq. ??) with electron beam temperatures $k_B T_{\perp} = T_{\text{PER}} = 1.47 \times 10^{-3}$ Ryd = 20 meV and $k_B T_{\parallel} = T_{\text{PAR}} = 1.47 \times 10^{-5}$ Ryd = 0.2 meV.
- Results are produced for NTAR1=2 initial levels, i.e., DR of $\text{Ar}^{14+}(1s^2 2s^2 {}^1S_0)$ and DR of $\text{Ar}^{14+}(1s^2 2s 2p {}^3P_0)$.
- NR1=3 specifies the lowest n outside the core for which DR is calculated.
- COREX='2-2' denotes that $\Delta n = 0$ core excitation within the L -shell is considered.
- The input below the namelist input is just a copy of the second to last line of the file Ar14/LEVELS calculated in section 2.1.
- The very last line contains NECOR=10 improved core-excitation energies (in Ryd, all on one line), e.g., from the NIST Atomic Spectra Database (<https://dx.doi.org/10.18434/T4W30F>).

Executing the calculation: Type "autosdr Ar14DR" on the command line.

Output files: The file Ar14DR.adr contains the cooler-convolved merged-beams rate coefficients (in $10^{-11} \text{ cm}^3/\text{s}$) as a function of electron energy (in Ryd). The file Ar14DR.bdr contains binned cross sections (in 10^{18} cm^2 , see also below) which can be processed further, e.g., by the HYDROCAL code.

2.4 Binned cross sections

The input file in section 2.3 defines an energy grid ranging from $E_{\min} = \text{EMIN}$ to $E_{\max} = \text{EMAX}$ containing $n_b = \text{NBIN} - 1$ energy bins of width $\Delta E = (E_{\max} - E_{\min})/n_b$. Bin number k ranges from $E_{\text{lo}}^{(k)} = E_{\min} + k \cdot \Delta E$ to $E_{\text{hi}}^{(k)} = E_{\text{lo}}^{(k)} + \Delta E$. For a Lorentzian resonance with resonance energy E_{res} , width Γ , and resonance strength $\bar{\sigma}$, i.e., for

$$\sigma(E) = \frac{\bar{\sigma}}{\pi} \frac{\Gamma/2}{(E - E_{\text{res}})^2 + (\Gamma/2)^2}. \quad (1)$$

the binned cross section for bin number k is

$$\sigma^{(k)} = \frac{\bar{\sigma}}{\pi \Delta E} \int_{E_{\text{lo}}^{(k)}}^{E_{\text{hi}}^{(k)}} \frac{\Gamma/2}{(E - E_{\text{res}})^2 + (\Gamma/2)^2} dE = \frac{\bar{\sigma}}{\pi \Delta E} \int_{\epsilon_{\text{lo}}^{(k)}}^{\epsilon_{\text{hi}}^{(k)}} \frac{1}{\epsilon^2 + 1} d\epsilon \quad (2)$$

where

$$\epsilon_{\text{lo}}^{(k)} = \frac{E_{\text{lo}}^{(k)} - E_{\text{res}}}{\Gamma/2} \quad \text{and} \quad \epsilon_{\text{hi}}^{(k)} = \frac{E_{\text{hi}}^{(k)} - E_{\text{res}}}{\Gamma/2}.$$

The integral can be easily solved analytically yielding

$$\sigma^{(k)} = \frac{\bar{\sigma}}{\pi \Delta E} \left[\arctan \left(\frac{E_{\text{hi}}^{(k)} - E_{\text{res}}}{\Gamma/2} \right) - \arctan \left(\frac{E_{\text{lo}}^{(k)} - E_{\text{res}}}{\Gamma/2} \right) \right]. \quad (3)$$

Internally AUTOSTRUCTURE uses this type of binning [2]. The convolution with the merged-beams energy distribution is carried out by representing the DR cross section as a sum of delta-like cross sections, i.e., by approximating $\alpha_{MB}(\hat{E})$ as

$$\alpha_{MB}(\hat{E}) = \sum_k \frac{(\sigma^{(k)} \Delta E) c \sqrt{E^{(k)}}}{k_B T_{\perp} \xi \sqrt{2m_e c^2}} \exp \left(-\frac{E^{(k)} - \hat{E}/\xi^2}{k_B T_{\perp}} \right) \times \left[\text{erf} \left(\frac{\sqrt{E^{(k)}} + \sqrt{\hat{E}/\xi^2}}{\sqrt{k_B T_{\parallel}}/\xi} \right) + \text{erf} \left(\frac{\sqrt{E^{(k)}} - \sqrt{\hat{E}/\xi^2}}{\sqrt{k_B T_{\parallel}}/\xi} \right) \right] \quad (4)$$

with $(\sigma^{(k)} \Delta E)$ from Eq. 3 and $E^{(k)} = (E_{\text{hi}}^k + E_{\text{lo}}^k)/2 = E_{\min} + (k + 0.5) \cdot \Delta E$

This approximation is sufficiently accurate as long as the bin width ΔE is much smaller than the experimental electron-energy spread [2]. It should however be kept in mind that the representation of DR resonances as Lorentzians is not always appropriate, in particular, not at low energies. Moreover, the resonance energies are calculated as differences of large numbers which bears large uncertainties again for low energy resonances. All these limitations lead to noticeable differences between AUTOSTRUCTURE results and experimental data, in particular, at low energies (see, e.g., [3, 4, 5, 6, 7, 8]).

3 RR calculations with AUTOSTRUCTURE

The recipe below produces some output, but its adequateness and its accuracy are currently still unclear. Please use with care!

AUTOSTRUCTURE has to calculate all atomic level energies and transition rates which are required for the computation of RR cross section. This is done partly by `asdeck.f` and partly (for high- n levels) by `adasrr.f`. The input files have to be named `<name>.in` and `<name>.rrin`, respectively, where `<name>` can be chosen freely. All files pertaining to a given calculation are either stored in an automatically created subdirectory `<name>` or are named `<name>.<ext>` with various extensions `<ext>` as detailed below. In the following, RR of O-like Ne^{2+} is treated as an example.

3.1 Levels energies

Level energies of the $\text{Ne}^{2+} 1s^2 2s^2 2p^4$, $1s^2 2s^1 2p^5$, and $1s^2 2p^6$ configurations.

Input file: `Ne2IC.in`

```
A.S. Ne2+
&SALGEB RUN=' ' CUP='IC' MXCONF=3 MXVORB=2 TITLE='Ne2+'
      KCOR1=1 KCOR2=1 &END
2 0 2 1
2 4
1 5
0 6
&SMINIM NZION=-10 PRINT='FORM' &END
```

See Section 2.1 for details.

Executing the calculation: Type "`autos Ne2IC`" on the command line.

Output file `Ne2IC/LEVELS:`

2J P	S L	CF	NI	ENERGY (RYD)
4 0	3 1	1	2	0.00000000
2 0	3 1	1	3	0.00609512
0 0	3 1	1	4	0.00896683

4	0	1	2	1	1	0.26198031
0	0	1	0	1	5	0.50851848
4	1	-3	1	2	1	2.01625991
2	1	-3	1	2	2	2.02226537
0	1	-3	1	2	4	2.02529861
2	1	-1	1	2	3	2.91469946
0	0	1	0	3	1	4.89144158
0	0	0	0	0	0	-253.01521551

The level energies are in units of Rydbergs (1 Ryd = 13.606 eV).

3.2 First step of RR calculation

RR of $\text{Ne}^{2+}(1s^2 2s^2 2p^4)$, capture into nlj -levels with $n \leq 15$.

Input file Ne2RR.in:

```
A.S. Ne2+RR
&SALGEB RUN='RR' RAD='E1' CUP='IC' MXVORB=2 MXCONF=3 TITLE='Ne2RR'
      KCOR1=1 KCOR2=1 LCON=3 &END
2 0 2 1
2 4
1 5
0 6
&DRR NMIN=3 NMAX=15 LMIN=0 LMAX=14 &END
&SMINIM NZION=-10 PRINT='FORM' MAXE=10 MSTEP=12 &END
&SRADCON EMAX=10 &END
```

Basically a simplified input from Section 2.2.

- Calculations are carried out for capture into nl levels for $NMIN \leq n \leq NMAX$ and $LMIN \leq l \leq LMAX$. Higher nl values are treated hydrogenically by the postprocessor (see Section 3.3 below).
- Note that MAXE and EMAX are used in the namelists SMINIM and SRADCON, respectively, for specifying the maximum electron energy in Rydberg.
- MSTEP specifies the number of energy steps.

Executing the calculation: Type "autos Ne2RR" on the command line.

Output files: The file Ne2RR/o1 contains intermediate output which is to be processed in step 2 below. If something goes wrong, the file Ne2RR/o1g may contain error messages.

3.3 Second step of RR calculation

Summation of cross section. For high n a hydrogenic approximation is used.

Input file Ne2RR.rrin:

```
/IC/
&ONE NTAR1=1 NTAR2=10 IPRINT=-3 &END
&TWO NR1=3 JTHETA=0 &END
 4 0   3 1   1   2   0.00000000
 2 0   3 1   1   3   0.00609512
 0 0   3 1   1   4   0.00896683
 4 0   1 2   1   1   0.26198031
 0 0   1 0   1   5   0.50851848
 4 1  -3 1   2   1   2.01625991
 2 1  -3 1   2   2   2.02226537
 0 1  -3 1   2   4   2.02529861
 2 1  -1 1   2   3   2.91469946
 0 0   1 0   3   1   4.89144158
 0 0   0 0   0   0  -253.01521551
```

Similar to input of Section 2.3. Further comments:

- IPRINT=-3 requests minimal output.
- NR1=3 specifies the lowest n outside the core for which DR is calculated.
- JTHETA=0 requests cross-section output.
- The input below the namelist input is just a copy of the second to last line of the file Ne2IC/LEVELS calculated in section 3.1. It contains NTAR2=10 levels.

Executing the calculation: Type "autosrr Ne2RR" on the command line.

Output files: The file Ne2RR.arr contains the RR cross section times the electron energy (in 10^{-18} cm² Ryd) as a function of electron energy (in Ryd).

References

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- [5] M. Lestinsky, N. R. Badnell, D. Bernhardt, D. Bing, M. Grieser, M. Hahn, J. Hoffmann, B. Jordon-Thaden, C. Krantz, O. Novotný, D. A. Orlov, R. Repnow, A. Shornikov, A. Müller, S. Schippers, A. Wolf, and D. W. Savin. “Electron-ion recombination of Mg^{6+} forming Mg^{5+} and of Mg^{7+} forming Mg^{6+} : laboratory measurements and theoretical calculations”. In: *Astrophys. J.* 758 (2012), p. 40. DOI: 10.1088/0004-637X/758/1/40.
- [6] M. Lestinsky, N. R. Badnell, D. Bernhardt, M. Grieser, J. Hoffmann, D. Lukić, E. W. Schmidt, A. Müller, D. A. Orlov, R. Repnow, D. W. Savin, S. Schippers, A. Wolf, and D. Yu. “Electron-ion recombination for Fe X forming Fe IX and of Fe XI forming Fe X: Laboratory measurements and theoretical calculations”. In: *Astrophys. J.* 698 (2009), pp. 648–659. DOI: 10.1088/0004-637X/698/1/648.
- [7] E. W. Schmidt, S. Schippers, D. Bernhardt, A. Müller, J. Hoffmann, M. Lestinsky, D. A. Orlov, A. Wolf, D. V. Lukić, D. W. Savin, and N. R. Badnell. “Electron-ion recombination for Fe VIII forming FeVII and FeIX forming FeVIII: measurements and theory”. In: *Astron. Astrophys.* 492 (2008), pp. 265–275. DOI: 10.1051/0004-6361:200810834.
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